Gravity on the 7+1 spacetime: provenance

Sources, decisions, derivations and the verification record of the curvature, the Einstein and Lovelock tensors, the generalized frame matrices, the canonical spin connection, the covariant Dirac equation of dirac16complex and the three cosmological epochs — sections 11–14 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states what was asked, what was used, every decision taken with the equation behind it, how each result was derived and how it was checked: the 100 checks of sections 11–14 of 7+1.nb (100 True), the 101 boolean checks of the independent Python program tools/ref_gravity.py (101 True), the comparison of the printed numbers, the program versions and the SHA-256 hashes of the files.

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1. The task

This stage of the project asked for the following (summarized; the original wording is the author's private working material and is not reproduced):

1. Take the IDEAS of the repository github.com/once-ere/Dirac_claude, which is cast on a 4+4 spacetime with four time-like coordinates and a definite warped metric, and rebuild them on the 7+1 spacetime of this project: ONE time-like coordinate x_0 and SEVEN space-like coordinates $x_1, \ldots, x_7$. The 4+4 metric of Dirac_claude is not used.
2. Einstein and Lovelock tensors of 7+1 spacetime.
3. Generalized frame matrices; a non-trivial canonical spin connection, which couples dirac16complex to gravity; and a total covariant derivative under which the frame matrices are constant.
4. Kohn–Sham theory of dirac16complex in gravitational fields (primordial near $x_0 = 0$, modern, far future), its self-coupling, and the totality of the Standard Model (sections 15–19 of 7+1.nb; this record covers sections 11–14).
5. Compute the Lovelock tensors as in the author's reference notebook Einstein-Lovelock-7+1-Nash.nb and do NOT use its section "FAILED Model metric g71; DO NOT USE!".

6. Make clear in all documentation, examples and provenance records that the xAct declaration $\{7,1,0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1,-1,\ldots,-1)$ is simply the squares of the octonion basis (the author's instruction of 27 October 2026).

7. For every documentation file a companion provenance file and a symbol file, each in Markdown and LaTeX, the LaTeX compiled to PDF; no document refers the reader to another one, and every equation used is written in the document that uses it.

8. Push every completed stage to github.com/once-ere/7plus1, respecting the repository's .gitignore; use only the Wolfram front end WolframNB.exe of Wolfram 15.0.1.

The coordinates of 7+1.nb are kept: $X = (x_0,\ldots,x_7)$, x_0 the time (the direction of the octonion unit), x_1,x_2,x_3 ordinary space, $x_4,\ldots,x_7$ auxiliary space.

1.1 The signature: the author's explanation of 27 October 2026

Source. On 27 October 2026 the author of the project explained that the xAct list $\{7,1,0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1,-1,\ldots,-1)$ is simply the squares of the octonion basis, and asked that this be made clear in all documentation, examples and provenance records. **What it corrected.** The version of 7+1.nb pushed in the commits 52d8729 to dc2f178 of github.com/once-ere/7plus1 declared the xAct metric of section 12 with `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of η; since then section 12 declares `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`, exactly as the author's notebook Einstein-Lovelock-7+1-Nash.nb does. **What was added.** Checks in 7+1.nb — section 4.7: η is the squares of the octonion basis, $e_A e_A = \eta_{AA} e_0$ and $\mathrm{Re}(e_A e_B) = \eta_{AB}$; section 12.2: `SignatureOfMetric[g71]` $= \{7,1,0\}$ with `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$, and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$; section 17: the flat metric $\mathrm{diag}(+1,-1,-1,-1)$ of the world brane is the squares of the basis e_0,e_1,e_2,e_3 of the quaternion subalgebra — and the same statements in the Python programs `tools/ref71.py`, `tools/ref_gravity.py` and `tools/ref_sm.py`; the explanation itself in the text of 7+1.nb (sections 1.2, 1.4, 4.2, 4.7, 7, 11, 12.2, 13, 15.1, 17) and in every document, symbol file and provenance file of the repository. **What did not change.** No result. Compared output by output with the backup below, every output of 7+1.nb is the same, except the declared list itself, the new checks and the check counts (and, as in every run, the name of the private context and the measured times); in particular every component of every Lovelock tensor is unchanged. The components follow from the component matrix of the metric, and the sign of the determinant, `SignDetOfMetric`, is -1 for both lists. **Backup.** The evaluated notebook as it was before is kept in the workspace as `backups/7+1_2026-10-27_before_signature.nb` (the folder backups/ is excluded from the repository by .gitignore), 1034401 bytes, SHA-256 `060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`; `tools/verify_7p1_nb.py` compares every input, text and section cell of 7+1.nb with it and finds every cell present, each one unchanged or changed by this explanation, and it compares every output in the same way.

The explanation, in full. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1,x_2,x_3 ordinary space, $x_4,\ldots,x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1,\ldots,e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same

in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

2. Inputs

2.1 Objects of the earlier sections of 7+1.nb

Sections 11–14 use, unchanged, the coordinates `X = {x0, …, x7}`, the metric `η17` $= \eta$, the identity and zero matrices `ID16`, `Zero16`, the check registry `check[name, test]`, and the real 16×16 Dirac matrices `Γ17[A]` of section 4 with

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}\ (i = 1,\dots,7), \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}1_{16}, \qquad \Gamma^A = \eta^{AA}\Gamma_A,$$

where the real 8×8 matrices τ_i^O represent the seven imaginary octonion units ($(\tau_i^O)^2 = -1_8$, $\tau_i^O\tau_j^O + \tau_j^O\tau_i^O = -2\delta_{ij}1_8$), and the Hamiltonian matrices `αD[j]` $= \Gamma^0\Gamma^j$, `βD` $= \Gamma_0$ of section 7. Before sections 11 and following were appended, the evaluated notebook was copied to `backups/7+1_2026-10-26_before_gravity.nb` (excluded from the repository by .gitignore).

```
backups/7+1_2026-10-26_before_gravity.nb
604743 bytes
SHA-256 438d972073e2fef9d7da6d57a541432b0fb1ea08424307f59ae2f3f7752bbfaf
```

2.2 The reference notebook Einstein-Lovelock-7+1-Nash.nb

The author's notebook Einstein-Lovelock-7+1-Nash.nb (P. L. Nash, 2022, GNU GPL; corrected version of 27 October 2026, with its PDF printout) shows how the Lovelock tensors are to be computed. It is the author's file; at the author's request it was added to the repository on 27 October 2026 (commit 730a263), with its PDF printout Einstein-Lovelock-7+1-Nash.pdf (until 28 October 2026 this record said that it is not part of the repository, which was no longer true after that commit). Taken from it, unchanged in substance:

- the xAct packages xCoba and xTras, the manifold `M8` with the abstract indices a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y, the chart `chartM8` with coordinate scalars `x0[]`, …, `x7[]`, and `MyArrayComponents`;
- the generalized Kronecker delta `kδ[lower_, upper_] := Det[Outer[delta, lower, upper]]`;
- the three Lovelock tensors with their normalization,

$$\mathrm{Lovelock}_k = -\frac{1}{2^{k+1}}\, R_{c_1d_1}{}^{f_1i_1}\cdots R_{c_kd_k}{}^{f_ki_k}\ \delta_{a\,f_1i_1\cdots f_ki_k}^{b\,c_1d_1\cdots c_kd_k} \quad (k=1,2,3), \qquad \mathrm{Lovelock}_1 = R_a{}^b - \tfrac{1}{2}\delta_a{}^b R,$$

- the evaluation of components by assigning a component metric and then setting the connection and the curvature tensors,

```
CD = LC[g]; RiemannCD = Riemann[CD]; RicciCD = Ricci[CD];
RicciScalarCD = RicciScalar[CD]; EinsteinCD = Einstein[CD];
```

- the form of the vacuum field equations and the procedure of solving them for w_2, w_3 (and Λ):

$$0 = -\Lambda\,\delta_\nu^\mu + H^{-2}w_1\,\mathrm{Lovelock}_1 + H^{-4}w_2\,\mathrm{Lovelock}_2 + H^{-6}w_3\,\mathrm{Lovelock}_3,$$

with the justification it quotes from D. Lovelock and H. Rund, *Tensors, Differential Forms, and Variational Principles* (Dover), Remark 2 after Eq. (4.38): in n dimensions the most general symmetric divergence-free tensor built from the metric and its first two derivatives is

$$A^{lh} = \sqrt{|g|}\sum_{k=1}^{m-1}\alpha_{(k)}\,g^{jl}\,\delta_{jj_1\cdots j_{2k}}^{hh_1\cdots h_{2k}}\,R^{j_1j_2}{}_{h_1h_2}\cdots R^{j_{2k-1}j_{2k}}{}_{h_{2k-1}h_{2k}} + \lambda\sqrt{|g|}\,g^{lh}, \qquad m-1 = \frac{8}{2} - 1 = 3\ \ (n=8).$$

NOT taken: the section "FAILED Model metric g71; DO NOT USE!" of the reference notebook (its model metric and the solution found for it); its author states that it never worked for the 7+1 signature. No metric of that section appears in this project. Taken unchanged as well: the signature declaration `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]` — in xAct seven space-like dimensions ($x_1, \ldots, x_7$), one time-like dimension (x_0), none degenerate — and the metric symbol `g71`.

A correction (27 October 2026; section 1.1). An earlier version of section 12 of 7+1.nb declared `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$; it now declares $\{7, 1, 0\}$, as the reference notebook does. The Lovelock tensors are the same with both declarations: they follow from the components of the metric, and `SignDetOfMetric` is -1 for both (a separate kernel with xTensor 1.2.0 confirmed this for a metric declared `{1, 7, 0}` and one declared `{7, 1, 0}`).

```
Einstein-Lovelock-7+1-Nash.nb   SHA-256 b2dcb7b9f435d644da085fdd9e738a0cc80d3268ef28606ff9f367ed
    ...6828fdfc
Einstein-Lovelock-7+1-Nash.pdf  SHA-256 6580f9fcaff9c7f1337df35a7fce6b4b04c022d0e2d68954e8b3954b
    ...30f45125
```

2.3 Software and other sources

- Wolfram Language 15.0.1 (the front end WolframNB.exe in `C:\Program Files\Wolfram Research\Wolfram\15.0.1`).
- xAct (J. M. Martín-García, D. Yllanes, T. Nutma and others; GPL; www.xact.es), versions as printed by section 12.2 of 7+1.nb: xPerm 1.2.3, xTensor 1.2.0, xCoba 0.8.6, xTras 1.4.2.
- Python 3.13.17 with sympy 1.14.0, numpy 2.5.3, scipy 1.18.1, mpmath 1.3.0 (the independent programs).
- Ideas (no formulas or metrics) from the repository github.com/once-ere/Dirac_claude: a 16-component field coupled to gravity by the canonical spin connection, Einstein–Lovelock gravity, a primordial pair-creation epoch, a Kohn–Sham treatment, the Standard Model on a lower-dimensional world surface.
- The author's private notes on vielbeins in n dimensions (Cartan's structure equation, a static metric with a radial coordinate, the Dirac equation, the Noether current and the U(1) coupling): their 7+1 versions were derived and checked anew (section 4 below gives the derivations, section 5 the checks). The notes are private and not part of the repository.
- Standard textbook results, each derived and checked here: the Landau–Lifshitz curvature conventions, the Lanczos tensor, Kasner solutions, the Gibbons–Hawking temperature $H/(2\pi)$, the Landau–Zener formula.

3. Decisions

#	decision	reason / equation
1	Christoffel symbols are written $\{^{\lambda}_{\mu\nu}\}$	Γ is reserved for the Dirac matrices
2	Riemann tensor of Landau and Lifshitz: $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\{^{\rho}_{\sigma\nu}\} - \partial_\nu\{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\}$, $R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}$	with the signs of η ($g_{00} > 0$): $G^{\mu}{}_{\nu} - \bar{\Lambda}\delta^{\mu}_{\nu} = \kappa T^{\mu}{}_{\nu}$ with $T^{0}{}_{0} > 0$; agrees with xAct (the reference notebook's $\mathrm{Lovelock}_1$ equals this G)
3	Lovelock tensors normalized as in the reference notebook, $E^{(k)} = -2^{-(k+1)}\delta RR\cdots$	$E^{(1)} = G$, $E^{(2)}$ = Lanczos tensor
4	two forms of the field equations: w-form (reference notebook) and α-form	$\alpha_2 = w_2/(w_1H^2)$, $\alpha_3 = w_3/(w_1H^4)$, $\bar{\Lambda} = \Lambda H^2/w_1$
5	xAct signature declaration $\{7, 1, 0\}$, as in the reference notebook	seven space-like dimensions and one time-like dimension; $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ is a different object, simply the squares of the octonion basis, $\eta_{AB} = \mathrm{Re}(e_A e_B)$
6	xAct runs in the notebook kernel; the cell is guarded (`If[! MemberQ[$Manifolds, M8], …]`, `Unset` of the component assignments); `tools/run_eval.wl` evaluates the notebook in a fresh kernel	the cell can be evaluated twice while the notebook stays open; xAct keeps its definitions in the kernel, while a reopened notebook gets a new private context

#	decision	reason / equation
7	`xActLovelock` returns the Transpose of `MyArrayComponents`	the array has the lower index a first; the pp-wave ($E^0{}_1 = -E^1{}_0$) fixes the order
8	a second, independent computation (pair-set sums) is kept	exact agreement with xAct is a strong check; it also works on exact rational tables
9	the inhomogeneous metric is an explicit polynomial, checked exactly at $x_0 = 1/2$, $x_4 = 1/3$	symbolic checks with arbitrary functions of two variables took hours
10	exponential expansion is evaluated from the xAct components of the cosmological metric (`xlCos`), not by a new xAct run	xAct's final Simplify with symbolic exponentials is very slow
11	the spinor connection is built as `Zero16 + Sum[…]`	a sum of zeros would otherwise be the scalar 0, not a 16×16 matrix
12	the 16-component mode equation is reduced with $\mathcal{Z} = \mathcal{A}\mathcal{B}\beta$, $\mathcal{Z}^2 = -1$	two classes of states ($\pm m\sigma_3$), created at different rates
13	pair creation is integrated from $10^{-15}\lambda$ to 300λ (Kasner), from $-\ln 2000$ to $\ln(10^6/M^2)$ (de Sitter)	adiabatic before and after; checked by $\lvert\alpha\rvert^2 + \lvert\beta\rvert^2 = 1$ and the Landau–Zener formula

4. Derivations

Sectional curvatures of the cosmological metric. In the frame $e = \mathrm{diag}(1, a, a, a, b, b, b, b)$ every $R^{AB}{}_{CD}$ is diagonal on index pairs, with

$$K_{0i} = -(H_a' + H_a^2), \quad K_{0k} = -(H_b' + H_b^2), \quad K_{ij} = -H_a^2, \quad K_{kl} = -H_b^2, \quad K_{ik} = -H_a H_b,$$

In $L_k = 2^{-k}\delta^{\alpha_1\beta_1\cdots}_{\gamma_1\delta_1\cdots} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots$ every ordered sequence of $2k$ different indices contributes $2^k\prod_i K_{\alpha_i\beta_i}$ (each lower pair is the upper pair in one of its two orders), so L_k is a sum over SETS of k disjoint index pairs, and $E^{(k)\mu}{}_\mu$ (no sum) is $-\frac{1}{2}$ times the same sum over the pairs that avoid μ:

$$L_k = 2^k k! \sum_{\{p_1,\ldots,p_k\}} K_{p_1}\cdots K_{p_k}, \qquad E^{(k)\mu}{}_\mu = -2^{k-1}k! \sum_{\{p_1,\ldots,p_k\}\not\ni\mu} K_{p_1}\cdots K_{p_k}.$$

Counting the pairs among $x_1,\ldots,x_7$ (3 inside ordinary space, 6 inside the auxiliary space, 12 mixed) and the sets of two disjoint pairs (18 of type ordinary–auxiliary, 12 ordinary–mixed, 3 auxiliary–auxiliary, 36 auxiliary–mixed, 36 mixed–mixed) gives

$$G^0{}_0 = -\big(3K_{12} + 6K_{45} + 12K_{14}\big) = 3H_a^2 + 12H_aH_b + 6H_b^2,$$
$$E^{(2)0}{}_0 = -4\big(12H_a^3H_b + 54H_a^2H_b^2 + 36H_aH_b^3 + 3H_b^4\big) = -12H_b\big(4H_a^3 + 18H_a^2H_b + 12H_aH_b^2 + H_b^3\big).$$

Kasner exponents. With $a = x_0^p$, $b = x_0^q$: $H_a = p/x_0$, $H_a' = -p/x_0^2$ (likewise for b), and $x_0^{2k}E^{(k)}$ is a polynomial in p, q. Einstein: $3p + 4q = 1$, $3p^2 + 4q^2 = 1$. Gauss–Bonnet: $3p + 4q = 3$ and $-203p^3 + 81p^2 + 351p + 27 = 0$. Cubic: $3p + 4q = 5$ and $(p, q) = (-15/7, 20/7)$, besides the degenerate families $q = 0$ and $p = 0$.

Spin connection of the cosmological metric. From $\omega_\mu{}^A{}_B = e^A{}_\nu(\partial_\mu e_B{}^\nu + \{{}^\nu_{\mu\lambda}\}e_B{}^\lambda)$ with $\{{}^0_{ii}\} = aa'$, $\{{}^i_{0i}\} = a'/a$: $\omega_i{}^0{}_i = a'$, hence $\Omega_i = \frac{1}{4}(\omega_{i0i}\Gamma^0\Gamma^i + \omega_{ii0}\Gamma^i\Gamma^0) = \frac{a'}{2}\Gamma_i\Gamma_0$.

Static radial metric (Cartan). $e^0 = \sqrt{f}dx_0$, $e^1 = \sqrt{h}dr$, $e^k = r dx^k$; $de^0 = \frac{f'}{2\sqrt{f}}dr\wedge dx_0$, $de^k = dr\wedge dx^k$, $de^1 = 0$; Cartan's equation $de^A + \omega^A{}_B\wedge e^B = 0$ gives

$$\omega^0{}_1 = \frac{f'}{2\sqrt{fh}}\,dx_0, \quad \omega^k{}_1 = \frac{dx^k}{\sqrt{h}}, \quad \Omega_0 = \frac{f'}{4\sqrt{f}\sqrt{h}}\Gamma_1\Gamma_0, \quad \Omega_k = \frac{1}{2\sqrt{h}}\Gamma_1\Gamma_k, \quad \Gamma^\mu\Omega_\mu = \frac{1}{\sqrt{h}}\Big(\frac{f'}{4f} + \frac{3}{r}\Big)\Gamma^1.$$

Exponential vacuum solutions. For $a = e^{Hx_0}$, $b = e^{q_H Hx_0}$ the vacuum equations are linear in (w_2, w_3, Λ), and the divergence identity $\frac{d}{dx_0}E^{(k)0}{}_0 + 3H_a(E^{(k)0}{}_0 - E^{(k)1}{}_1) + 4H_b(E^{(k)0}{}_0 - E^{(k)4}{}_4) = 0$ with constant entries makes them of rank 2. For $q_H = -1$: $w_2 = \Lambda/4 - w_1/2$, $w_3 = w_1/24 - \Lambda/36$.

de Sitter pair creation. For $i\dot{c} = (ke^{-Hx_0}\sigma_1 + M\sigma_3)c$ the change of variable $z \propto e^{-Hx_0}$ gives Whittaker functions; their connection formula yields $|\beta|^2 = 1/(e^{2\pi M/H} + 1)$, checked numerically below.

5. Verification

Evaluation of 7+1.nb. The notebook was generated by `tools/make_7p1_nb.py` (which executes `tools/nb_part_gravity.py`, `tools/nb_part_ks_gravity.py`, `tools/nb_part_sm.py`) and evaluated by `tools/run_eval.wl` in a fresh kernel of Wolfram 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026); seconds 283.883, message cells 0, finished Wed 28 Oct 2026 08:57:47. The final section of the notebook lists 479 checks; all passed: True.

The checks of sections 11–14 of 7+1.nb:

#	check	result	section
1	cosmological metric: sectional curvatures K_0i = -(Ha'+Ha^2), K_0k = -(Hb'+Hb^2), K_ij = -Ha^2, K_kl = -Hb^2, K_ik = -Ha Hb	True	11.2
2	cosmological metric: R^cd_ab is non-zero only for {c,d} = {a,b}	True	11.2
3	Einstein tensor: G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2, G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb, G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True	11.2
4	scalar curvature R = -2 (3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 Ha Hb)	True	11.2
5	inhomogeneous metric, exactly at (x0, x4) = (1/2, 1/3): R_rsmn = -R_srmn = -R_rsnm = R_mnrs and R^r_[smn] = 0 (first Bianchi identity)	True	11.3
6	inhomogeneous metric: contracted Bianchi identity nabla_mu G^mu_nu == 0, exactly at (x0, x4) = (1/2, 1/3)	True	11.3
7	cosmological metric: contracted Bianchi identity nabla_mu G^mu_nu = 0 (arbitrary a, b)	True	11.3
8	generalized Kronecker delta: contraction identity delta^{m1..mp r}_{n1..np r} = (8-p) delta^{m1..mp}_{n1..np} (p = 1..4, 60 random index sets each)	True	12.1
9	generalized Kronecker delta with 9 upper indices vanishes in 8 dimensions (100 random index sets)	True	12.1
10	xAct declaration as in the reference notebook: SignatureOfMetric[g71] == {7, 1, 0} (seven space-like dimensions, one time-like dimension, none degenerate) and SignDetOfMetric[g71] == -1	True	12.2
11	eta17 is simply the squares of the octonion basis: eta17[[A+1, B+1]] == Re(e_A e_B), e_0^2 == +1 (time x0), e_i^2 == -1 (space x_i, i = 1..7)	True	12.2
12	the quadratic form of the octonions has the signs of the squares of the octonion basis: Re(x x) == x0^2 - x1^2 - ... - x7^2 == x.eta17.x for x = x^A e_A (symbolic)	True	12.2
13	xAct: Lovelock1 == R_a^b - (1/2) delta_a^b R, the Einstein tensor (abstract indices)	True	12.3
14	xAct: trace of Lovelock2 == -2 L_2 with L_2 = R^2 - 4 R_cd R^cd + R_cdfi R^cdfi (abstract indices)	True	12.3
15	xAct, cosmological metric: Lovelock1 == EinsteinCD == the Einstein tensor of the curvature engine of section 11.1	True	12.4
16	pair-set algorithm == brute-force generalized-delta contraction for E^(1) (cosmological metric)	True	12.5
17	pair-set algorithm == xAct (section 12.4) for E^(1), E^(2), E^(3) of the cosmological metric (two independent computations)	True	12.5
18	E^(1) == Einstein tensor G and L_1 == R (cosmological metric)	True	12.6
19	trace identity E^(k)mu_mu == -((8-2k)/2) L_k for k = 0..4	True	12.6
20	E^(4) == 0 identically in 8 dimensions, while the Euler density L_4 != 0	True	12.6
21	every E^(k) is diagonal, E^(k)1_1 = E^(k)2_2 = E^(k)3_3 and E^(k)4_4 = ... = E^(k)7_7	True	12.6
22	E^(2) == Lanczos tensor and L_2 == R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True	12.6
23	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2 Hb + 12 Ha Hb^2 + Hb^3) and E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True	12.6
24	divergence-free: d0 E^0_0 + 3Ha (E^0_0 - E^1_1) + 4Hb (E^0_0 - E^4_4) == 0 for k = 1, 2, 3 (arbitrary a, b)	True	12.6
25	variation of nL a^3 b^4 L_k: EL_nL = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4 (k = 1..4)	True	12.6
26	inhomogeneous metric: nabla_mu E^(k)mu_nu == 0 EXACTLY at (x0, x4) = (1/2, 1/3) for k = 1, 2, 3	True	12.7
27	inhomogeneous metric: E^(1) == G at the point, and E^(k)_mu nu symmetric (k = 1, 2, 3)	True	12.7
28	inhomogeneous metric: E^(2) == Lanczos tensor at the point (exact)	True	12.7
29	pp-wave: E^(1)_mu nu == +(1/2)(d2^2 + d3^2) fPP l_mu l_nu	True	12.8
30	pp-wave: E^(2) == E^(3) == 0 for every profile fPP	True	12.8
31	pp-wave: xAct (section 12.4) == pair-set algorithm for E^(1), E^(2), E^(3) (a metric that is not diagonal; arbitrary profile fPP)	True	12.8
32	exponential expansion a = e^{H x0}, b = e^{qH H x0}: xAct Lovelock_k == the closed forms of section 12.6 with Ha = H, Hb = qH H, Ha' = Hb' = 0 (k = 1, 2, 3)	True	12.9
33	exponential expansion: three distinct vacuum equations, related by 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) == 0	True	12.9

#	check	result	section
34	exponential vacuum solutions: (w2, w3) solve all 64 components for every qH (generic qH) and every Lambda	True	12.9
35	eternal inflation (a = e^{H x0}) with deflation (b = e^{-H x0}): w2 == Lambda/4 - w1/2, w3 == w1/24 - Lambda/36; for Lambda = 0: w2/w1 == -1/2, w3/w1 == 1/24	True	12.9
36	cosmological metric: Omega_0 == 0, Omega_i == (a'/2) Gamma17[i].Gamma17[0], Omega_k == (b'/2) Gamma17[k].Gamma17[0]	True	13.2
37	cosmological metric: Gamma^mu Omega_mu == (3a'/(2a) + 2b'/b) Gamma17[0]	True	13.2
38	current identity: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi] (psi, psibar arbitrary)	True	13.2
39	Lichnerowicz formula: (Gamma^mu D_mu)^2 psi == (g^mu nu nabla_mu D_nu - R/4) psi	True	13.2
40	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi == (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True	13.2
41	mode equation: Gamma17[0].(i Gamma^mu D_mu - m) psi == (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} (i dc/dx0 - h(x0) c) for psi = (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} c(x0)	True	13.2
42	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) == C^(-9/2) Gamma17up[A] d_A f for g = C(x0)^2 eta17	True	13.4
43	Cartan's first structure equation de^A + omega^A_B ^ e^B == 0 (no torsion): cosmological, pp-wave and inhomogeneous frames	True	13.5
44	structure coefficients: omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA) with C^A_BC = e_B^mu e_C^nu (d_mu e^A_nu - d_nu e^A_mu): cosmological and inhomogeneous frames	True	13.5
45	static radial metric: Cartan's first structure equation and omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA)	True	13.5
46	static radial metric: Omega_0 == (f'/(4 Sqrt[f] Sqrt[h])) Gamma17[1].Gamma17[0], Omega_1 == 0, Omega_k == (1/(2 Sqrt[h])) Gamma17[1].Gamma17[k] (k = 2..7)	True	13.5
47	static radial metric: Gamma^mu Omega_mu == (1/Sqrt[h]) (f'/(4 f) + 3/r) Gamma17up[1]	True	13.5
48	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection, Gamma^mu D_mu psi == f^(-1/4) r^(-3) Gamma^mu d_mu chi	True	13.5
49	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) == e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A'_mu = A_mu + d_mu alpha	True	13.5
50	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True	13.5
51	Kasner metric: the Einstein tensor computed directly from the metric equals the closed form (x0^2 G^mu_mu, mu = 0, 1, 4)	True	14.1
52	Einstein Kasner: (p, q) = ((1+2Sqrt[2])/7, (2-3Sqrt[2])/14) and ((1-2Sqrt[2])/7, (2+3Sqrt[2])/14) solve E^(1) == 0	True	14.1
53	Gauss-Bonnet Kasner: 3 real roots of the cubic; (p, (3-3p)/4) solve E^(2) == 0	True	14.1
54	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) == 0; E^(3) == 0 identically for q = 0 and for p = 0	True	14.1
55	2x2 reduction: A, B, betaD square to ID16 and anticommute; Zc = A.B.betaD: Zc.Zc == -ID16, Transpose[Zc] == -Zc, Zc commutes with A, B, betaD	True	14.2
56	Kasner pair creation: |alpha|^2 + |beta|^2 == 1 (to 10^-8) for (|p|, |q|) = (1,1), (2,1), (1,3)	True	14.3
57	Kasner scaling: (|p|, |q|) = (1.4302.., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 10^-6)	True	14.3
58	16x16 evolution == 4 copies of the 2x2 evolution with +m sigma3 and 4 with -m sigma3 (Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0 from 0.05 to 40): average |beta|^2 agrees to 10^-8; unitarity; the two classes differ	True	14.3
59	isotropic expansion, massless: no pair creation (|beta|^2 < 10^-20)	True	14.3
60	static auxiliary space (Hb = 0): E^(1) = (3Ha^2, 2Ha' + 3Ha^2, 3Ha' + 6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha' + Ha^2)), E^(3) = 0	True	14.4
61	8D de Sitter: E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (all diagonal entries)	True	14.5
62	de Sitter over a static auxiliary space: E^(1) = (3, 3, 6) H^2, E^(2) = (0, 0, -12) H^4, E^(3) = 0	True	14.5
63	de Sitter pair creation: |beta|^2 == 1/(Exp[2 Pi M/H] + 1) to 10^-7, M/H = 0.25, 0.5, 1	True	14.5
64	integrator benchmark (Landau-Zener): |beta|^2 == Exp[-Pi kappa^2] to 10^-6	True	14.5
65	cosmological metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
66	cosmological metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
67	cosmological metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
68	cosmological metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
69	cosmological metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1

#	check	result	section
70	cosmological metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
71	cosmological metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
72	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
73	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
74	pp-wave: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
75	pp-wave: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
76	pp-wave: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
77	pp-wave: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
78	pp-wave: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
79	pp-wave: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
80	pp-wave: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
81	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
82	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
83	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
84	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
85	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
86	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
87	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
88	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
89	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
90	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
91	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1
92	static radial metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True	13.1
93	static radial metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True	13.1
94	static radial metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True	13.1
95	static radial metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True	13.1
96	static radial metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True	13.1
97	static radial metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True	13.1
98	static radial metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True	13.1
99	static radial metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True	13.1
100	static radial metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True	13.1

The independent Python program tools/ref_gravity.py (its own curvature and Lovelock engine tools/lovelock71.py; the Dirac matrices rebuilt from the split octonions of 4+4.nb by tools/ref71.py). Its boolean checks:

#	check	result
1	eta17 is simply the squares of the octonion basis: eta_AB = Re(e_A e_B), e_0^2 = +1, e_i^2 = -1 (i = 1..7)	True
2	the quadratic form of the octonions has the signs of the squares of the octonion basis: Re(x x) = x0^2 - x1^2 - ... - x7^2 = x.eta17.x	True

#	check	result
3	cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2, -Ha Hb	True
4	cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b}	True
5	G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2	True
6	G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb	True
7	G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True
8	scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb)	True
9	E^(1) = Einstein tensor G (cosmological metric)	True
10	L_1 = R	True
11	E^(0) = -delta/2	True
12	trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4	True
13	E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0	True
14	E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ... = E^(k)7_7	True
15	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3)	True
16	E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True
17	E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2, and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True
18	divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b)	True
19	variation of sqrt|g| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4	True
20	inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point	True
21	inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3	True
22	inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3	True
23	pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1	True
24	pp-wave: E^(2) = E^(3) = 0 for every profile F	True
25	Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2 = 1	True
26	Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3	True
27	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate families q = 0 and p = 0	True
28	Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0)	True
29	8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times delta)	True
30	4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) = diag(0,0,-12) H^4, E^(3) = 0	True
31	static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha'+Ha^2)), E^(3) = 0	True
32	exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha = H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3)	True
33	exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three vacuum equations have rank 2)	True
34	exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all three equations	True
35	eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 - Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24)	True
36	cosmological metric: eta e e = g	True
37	cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
38	cosmological metric: Gamma_mu real	True
39	cosmological metric: omega_mu AB = -omega_mu BA	True
40	cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
41	cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
42	cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
43	cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
44	cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
45	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
46	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
47	cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0]	True
48	cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2) Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0	True
49	pp-wave: eta e e = g	True
50	pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
51	pp-wave: Gamma_mu real	True
52	pp-wave: omega_mu AB = -omega_mu BA	True

#	check	result
53	pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
54	pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
55	pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
56	pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
57	pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
58	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
59	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
60	inhomogeneous metric N, A, B (x0, x4): eta e e = g	True
61	inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
62	inhomogeneous metric N, A, B (x0, x4): Gamma_mu real	True
63	inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA	True
64	inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
65	inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
66	inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
67	inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
68	inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
69	inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
70	inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
71	Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the cosmological frame	True
72	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g	True
73	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} = 2 g_mu nu	True
74	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real	True
75	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB = -omega_mu BA	True
76	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0	True
77	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu	True
78	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0	True
79	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0	True
80	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B	True
81	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0	True
82	static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16	True
83	Cartan's first structure equation for the static radial metric	True
84	structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) = -(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric)	True
85	static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k = (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7)	True
86	static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) + 3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi	True
87	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi = f^(-1/4) r^(-3) Gamma^mu d_mu chi	True
88	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A' = A + d alpha	True
89	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True
90	current identity d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi]	True
91	Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi	True
92	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True

#	check	result
93	mode equation: (i Gamma^mu D_mu - m) psi = 0 <=> i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) + beta m	True
94	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta	True
95	2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta	True
96	Kasner: |alpha|^2 + |beta|^2 = 1 (to 1e-8)	True
97	Kasner scaling: (|p|, |q|) = (1.430..., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 1e-6)	True
98	16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3: average agrees (to 1e-8), unitarity; the two sectors differ	True
99	isotropic expansion, massless: no pair creation (|beta|^2 < 1e-20)	True
100	de Sitter: |beta|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature H/2pi), to 1e-7	True
101	integrator benchmark: Landau-Zener |beta|^2 = exp(-pi kappa^2) (to 1e-6)	True

Numbers computed by both programs:

quantity	7+1.nb	Python
Kasner $\|\beta\|^2$ at $(\|\vec{p}\|, \|\vec{q}\|) = (1, 1)$	0.010749346	0.010749346
Kasner $\|\beta\|^2$ at $(\|\vec{p}\|, \|\vec{q}\|) = (2, 1)$	0.00013294108	0.00013294108
Kasner $\|\beta\|^2$ at $(\|\vec{p}\|, \|\vec{q}\|) = (1, 3)$	0.067442022	0.067442022
de Sitter $\|\beta\|^2$, $M/H = 0.25$	0.172102919	—
de Sitter $\|\beta\|^2$, $M/H = 0.5$	0.0414238845	—
de Sitter $\|\beta\|^2$, $M/H = 1$	0.00186397746	—

SHA-256 of the files (at the time this document was generated):

```
7+1.nb                          41d252f9852f210c228f45e1ac956d2d7ef26faeb0062a5880280e1aae0ef
    ...fc7
tools/make_7p1_nb.py            996d26c9552cd44902560af16a1683fa71457261f18fefaeddae87bd99a2b
    ...144
tools/nb_part_gravity.py        c4caf2ad5c932e8f603341edade5f4b98fa784feb05906c051eab7c884cd6
    ...8f8
tools/nb_part_ks_gravity.py     a656c0ad040b4576b75b17e029545109e531c8e656ad99bfdb5cba946e967
    ...772
tools/nb_part_sm.py             43e92beed57453a4b57931de6d603ac6c218efba6e4005133b69b8054a975
    ...c5b
tools/nb_part_coupled.py        e630521d64132e71afc861114f699eb06454557cf14a57ad8ec1eef96247a
    ...ae0
tools/nb_part_dark.py           10ad8fb8ae25fa07bbee758c653995acd4e2c80733059a8b3648bd1593e46
    ...db0
tools/nb_part_torus_brane.py    fb904ff8d85b904c55b3b2a8f60fb0cb047f1ccf1b98db3364092f2bae0b2
    ...162
tools/nb_part_radion.py         53c5604707a3e22228c840ee0a2d2125616f9410f75a5535041dfaf18f497
    ...c00
tools/lovelock71.py             d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af9
    ...5a1
tools/ref_gravity.py            f07cf8adaff585af89c87d19532f1aaa7db606ad88ae90483b2d1f41341ed
    ...f36
tools/ref_ks_gravity.py         b6488ef24b1c16aacd8ff6cf6a5a1ba3f316e082449ad97480022d1a26383
    ...27f
tools/ref_sm.py                 f3f80806893f42bbf166e1f1de896ffd611dbfe847685da8e3ee367e9cde6
    ...aba
tools/ref_coupled.py            695dbd9375753653472d9c37502f0067c4a64fe01a64ba8c05e8b2c5da85a
    ...a58
tools/ref_dark.py               1600e6b57ab0383b4f2b664e2796d3864c9afa77a9ed59f007f66e1139453
    ...ff5
tools/ref_torus_brane.py        1f623656ab0f5c8ee43cb310f7bfd580cbba5a7385da2963adaae4d2f8bda
    ...af7
```

```
tools/ref_radion.py                e9d47176b9f2d2232a84b3b82188bd3b0a19cde3dbe1ba013c1fe5abff634
    ...1ce
tools/ref71.py                     1a7a06eb0add3f459611776dc9c474a5bb959d947a9db949db4612b1ab943
    ...4e0
tools/ref44.py                     ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064f
    ...af5
```

Output of tools/ref_gravity.py (independent checks of sections 11-14; exit code 0) (complete):

```
eta17 is simply the squares of the octonion basis: eta_AB = Re(e_A e_B), e_0^2 = +1, e_i^2 = -1
    ...(i = 1..7) : True
the quadratic form of the octonions has the signs of the squares of the octonion basis: Re(x x)
    ...= x0^2 - x1^2 - ... - x7^2 = x.eta17.x : True
cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2,
    ...-Ha Hb : True
cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b} : True
G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2 : True
G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb : True
G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb : True
scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb) : True
E^(1) = Einstein tensor G (cosmological metric) : True
L_1 = R : True
E^(0) = -delta/2 : True
trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4 : True
E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0 : True
E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ...
    ...= E^(k)7_7 : True
E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3) : True
E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb) : True
E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2,
    ...and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric) : True
closed forms : {(1, 0): 3*H_a**2 + 12*H_a*H_b + 6*H_b**2, (1, 1): 3*H_a**2 + 8*H_a*H_b +
    ...10*H_b**2 + 2*dH_a + 4*dH_b, (1, 4): 6*H_a**2 + 9*H_a*H_b + 6*H_b**2 + 3*dH_a + 3*dH_b,
    ...(2, 0): -48*H_a**3*H_b - 216*H_a**2*H_b**2 - 144*H_a*H_b**3 - 12*H_b**4, (2, 1):
    ...-32*H_a**3*H_b - 136*H_a**2*H_b**2 - 16*H_a**2*dH_b - 192*H_a*H_b**3 - 32*H_a*H_b*dH_a -
    ...96*H_a*H_b*dH_b - 60*H_b**4 - 48*H_b**2*dH_a - 48*H_b**2*dH_b, (2, 4): -12*H_a**4 -
    ...108*H_a**3*H_b - 180*H_a**2*H_b**2 - 12*H_a**2*dH_a - 36*H_a**2*dH_b - 108*H_a*H_b**3 -
    ...72*H_a*H_b*dH_a - 72*H_a*H_b*dH_b - 12*H_b**4 - 36*H_b**2*dH_a - 12*H_b**2*dH_b, (3, 0):
    ...1440*H_a**3*H_b**3 + 1080*H_a**2*H_b**4, (3, 1): 576*H_a**3*H_b**3 + 1368*H_a**2*H_b**4
    ...+ 864*H_a**2*H_b**2*dH_b + 576*H_a*H_b**5 + 576*H_a*H_b**3*dH_a + 576*H_a*H_b**3*dH_b +
    ...144*H_b**4*dH_a, (3, 4): 648*H_a**4*H_b**2 + 1224*H_a**3*H_b**3 + 432*H_a**3*H_b*dH_b +
    ...648*H_a**2*H_b**4 + 648*H_a**2*H_b**2*dH_a + 648*H_a**2*H_b**2*dH_b +
    ...432*H_a*H_b**3*dH_a}
divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b) : True
variation of sqrt|g| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4
    ...E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4 : True
inhomogeneous metric: E^(3)0_4 at the point (exact) :
    ...353743050254866364332989413672793316453216419840000/24577736888913221711303629009057562607
    ...63178144678501
inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point : True
inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3 : True
inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3 : True
pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1 : True
pp-wave: E^(2) = E^(3) = 0 for every profile F : True
Kasner E^(k) components x0^(2k) (00, 11, 44) : {1: [3*(p**2 + 4*p*q + 2*q**2), 3*p**2 + 8*p*q -
    ...2*p + 10*q**2 - 4*q, 3*(2*p**2 + 3*p*q - p + 2*q**2 - q)], 2: [-12*q*(4*p**3 + 18*p**2*q
    ...+ 12*p*q**2 + q**3), -4*q*(8*p**3 + 34*p**2*q - 12*p**2 + 48*p*q**2 - 36*p*q + 15*q**3 -
    ...12*q**2), -12*(p**4 + 9*p**3*q - p**3 + 15*p**2*q**2 - 9*p**2*q + 9*p*q**3 - 9*p*q**2 +
    ...q**4 - q**3)], 3: [360*p**2*q**3*(4*p + 3*q), 72*p*q**3*(8*p**2 + 19*p*q - 20*p + 8*q**2
    ...- 10*q), 72*p**2*q**2*(9*p**2 + 17*p*q - 15*p + 9*q**2 - 15*q)]}
Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2
    ...= 1 : True
Gauss-Bonnet Kasner cubic (64 x e4 at q = (3-3p)/4) : -203*p**3 + 81*p**2 + 351*p + 27
```

```
Gauss-Bonnet Kasner exponents p (three real roots) : (-1.083501787612105742423976294 73,
    ...-0.078631058418927766227861052109 9, 1.561147624356156661361196952 75)
Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3 : True
cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate
    ...families q = 0 and p = 0 : True
Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0) : True
8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times
    ...delta) : True
4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) =
    ...diag(0,0,-12) H^4, E^(3) = 0 : True
static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2
    ...(Ha'+Ha^2)), E^(3) = 0 : True
exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha =
    ...H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3) : True
exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three
    ...vacuum equations have rank 2) : True
exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all
    ...three equations : True
eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 -
    ...Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24) : True
NUM exponential vacuum solution (w2, w3) for general q : ((-2*Lambda*q - 3*Lambda + 12*q**3*w1
    ...+ 27*q**2*w1 + 22*q*w1 + 9*w1)/(4*q*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)),
    ...(-3*Lambda*q**2 - 6*Lambda*q - Lambda + 15*q**4*w1 + 36*q**3*w1 + 33*q**2*w1 + 18*q*w1 +
    ...3*w1)/(72*q**3*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)))
NUM eq00, eq11, eq44 for q = -1 : (-Lambda - 3*w1 - 36*w2 - 360*w3, -Lambda + 5*w1 + 28*w2 +
    ...216*w3, -Lambda + 3*w1 + 12*w2 + 72*w3)
cosmological metric: eta e e = g : True
cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
cosmological metric: Gamma_mu real : True
cosmological metric: omega_mu AB = -omega_mu BA : True
cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu =
    ...0 : True
cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu
    ...nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 :
    ...True
cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0] : True
cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2)
    ...Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0 : True
pp-wave: eta e e = g : True
pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
pp-wave: Gamma_mu real : True
pp-wave: omega_mu AB = -omega_mu BA : True
pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l
    ...+ [Omega_mu, Gamma_nu] = 0 : True
pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
inhomogeneous metric N, A, B (x0, x4): eta e e = g : True
inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
inhomogeneous metric N, A, B (x0, x4): Gamma_mu real : True
inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA : True
inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l +
    ...omega_mu^A_B e^B_nu = 0 : True
inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] =
    ...-Omega_mu : True
inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu
    ...Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
```

```
inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B +
    ...[Omega_mu, Gamma^A] = 0 : True
inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A
    ...Gamma^B : True
inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0
    ...: True
inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the
    ...cosmological frame : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} =
    ...2 g_mu nu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB =
    ...-omega_mu BA : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate
    ...d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real,
    ...Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant
    ...derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu,
    ...Gamma_nu] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame
    ...matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu
    ...= (1/4) R_AB mu nu Gamma^A Gamma^B : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l
    ...F_mn + [Omega_l, F_mn] + cyclic = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu
    ...Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation for the static radial metric : True
structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) =
    ...-(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric) : True
static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k
    ...= (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7) : True
static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i
    ...Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) +
    ...3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi : True
static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi
    ...= f^(-1/4) r^(-3) Gamma^mu d_mu chi : True
U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu
    ...(D_mu - i q A_mu) psi with A' = A + d alpha : True
current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu
    ...psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu : True
current identity d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi +
    ...psibar Gamma^mu D_mu psi] : True
Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi : True
rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3
    ...b^4)^(-1/2) Gamma^mu d_mu chi : True
mode equation: (i Gamma^mu D_mu - m) psi = 0  <=>  i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) +
    ...beta m : True
conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta :
    ...True
2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B
    ...beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta : True
Kasner (Einstein branch), massless: |beta|^2 for (|p|, |q|) = (1,1), (2,1), (1,3) :
    ...(np.float64(0.01074934616777871), np.float64(0.00013294108220674155),
    ...np.float64(0.06744202228393215))
Kasner scaling variable K = |p| lambda^(1-p), lambda = (|q|/|p|)^(1/(q-p)) : (1.0,
    ...3.11828865465698, 0.4946322370571032)
Kasner: |alpha|^2 + |beta|^2 = 1 (to 1e-8) : True
Kasner scaling: (|p|, |q|) = (1.430..., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to
    ...1e-6) : True
Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0: 0.05 -> 40: |beta|^2 of the 16x16 evolution (average),
    ...of the sector Z = +i (+m sigma3) and of the sector Z = -i (-m sigma3) :
```

```
    ...(np.float64(0.027052647156543617), np.float64(0.02528659275098409),
    ...np.float64(0.028818701562187667))
16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3:
    ...average agrees (to 1e-8), unitarity; the two sectors differ : True
isotropic expansion, massless: no pair creation (|beta|^2 < 1e-20) : True
far future (de Sitter, H = 1): |beta|^2 for m/H = 0.25, 0.5, 1 :
    ...(np.float64(0.1721029191835808), np.float64(0.04142388447252214),
    ...np.float64(0.001863977463333981))
de Sitter: |beta|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature
    ...H/2pi), to 1e-7 : True
integrator benchmark: Landau-Zener |beta|^2 = exp(-pi kappa^2) (to 1e-6) : True
boolean checks: 101   failed: 0
VERDICT: ALL CHECKS PASSED
```